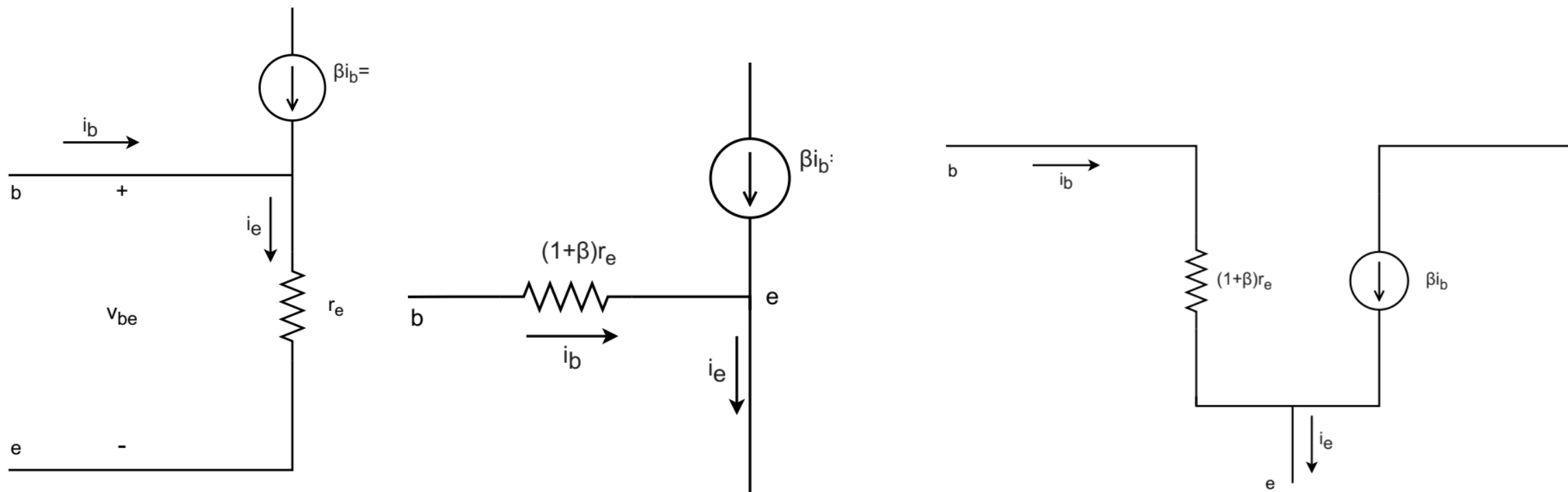
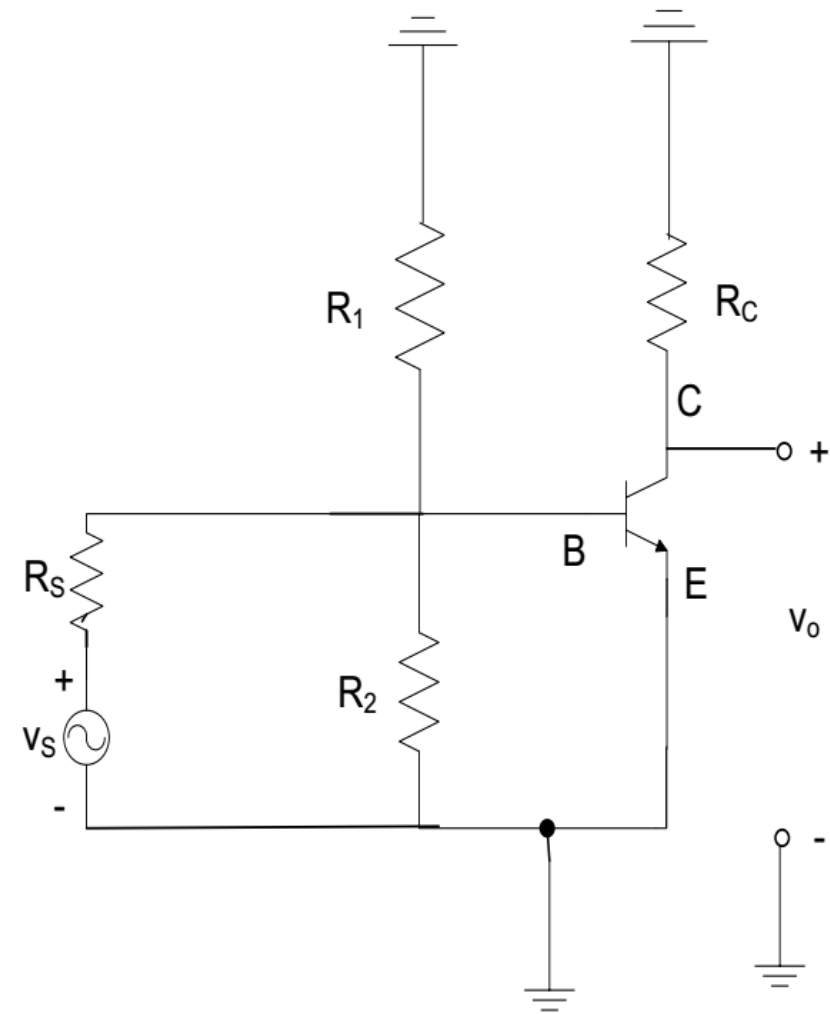
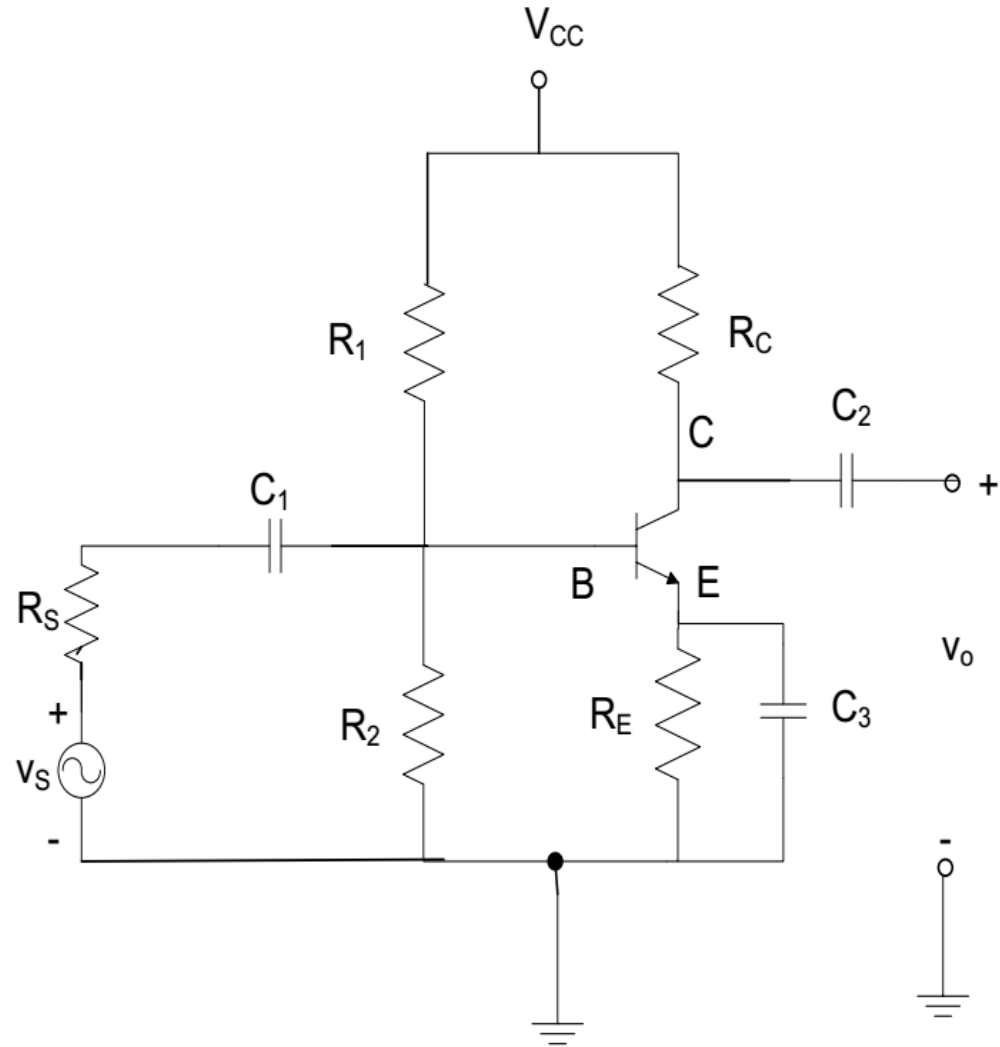




$$\begin{aligned}
 i_e r_e &= (\beta + 1) i_b r_e \\
 &= i_b (\beta + 1) r_e = i_b r_b
 \end{aligned}$$

A.C. Equivalent Circuit is obtained by :

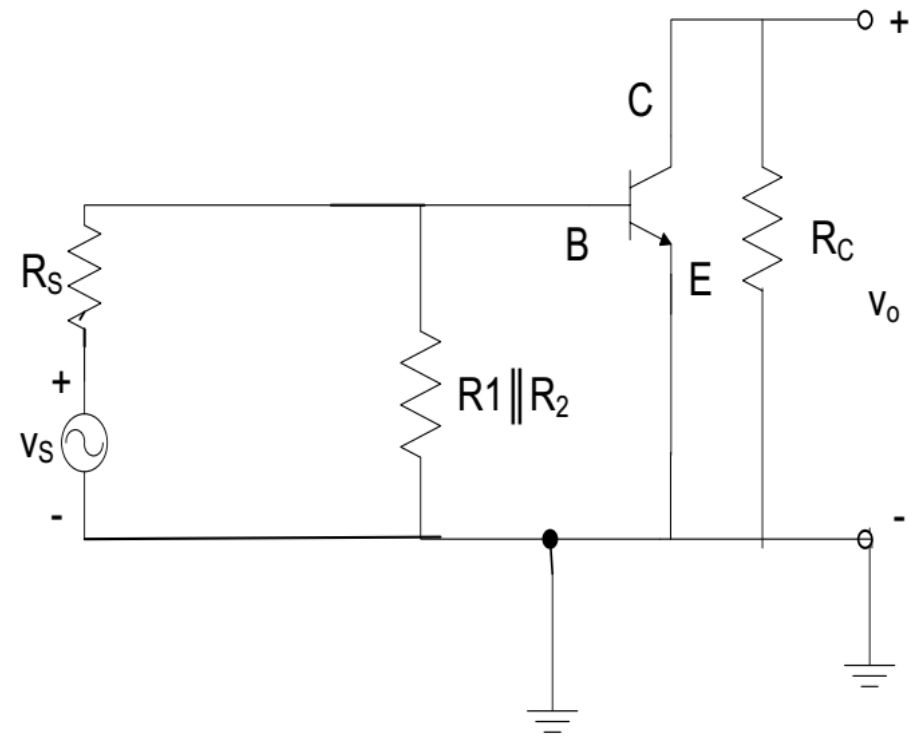
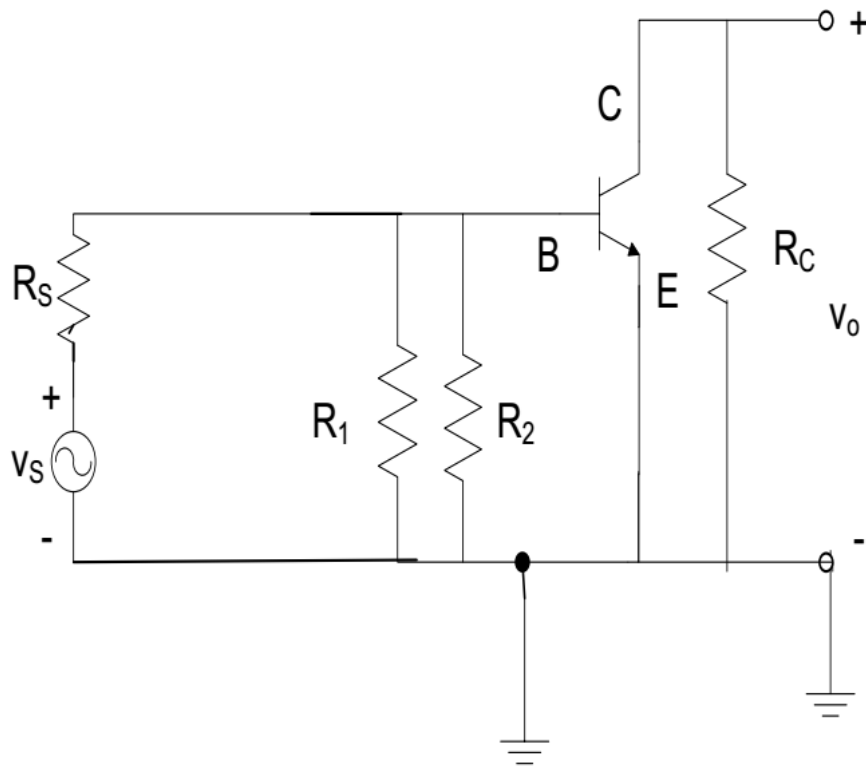
1. Setting all D.C. sources to 0 and replacing them by a short circuit equivalent
2. Replacing all capacitors by a short circuit equivalent
3. Removing all elements bypassed by the short circuit equivalents introduced in steps 1 and 2
4. Redrawing the network in a more convenient and logical form



Basic BJT Amplifier



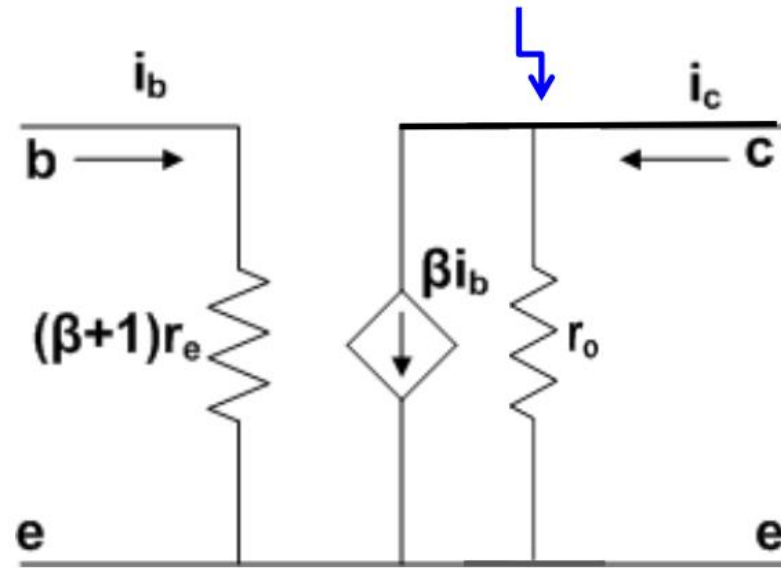
A.C. Equivalent Circuit



Simplify circuit by replacing R_1 and R_2 with R_B $R_B = R_1 || R_2$

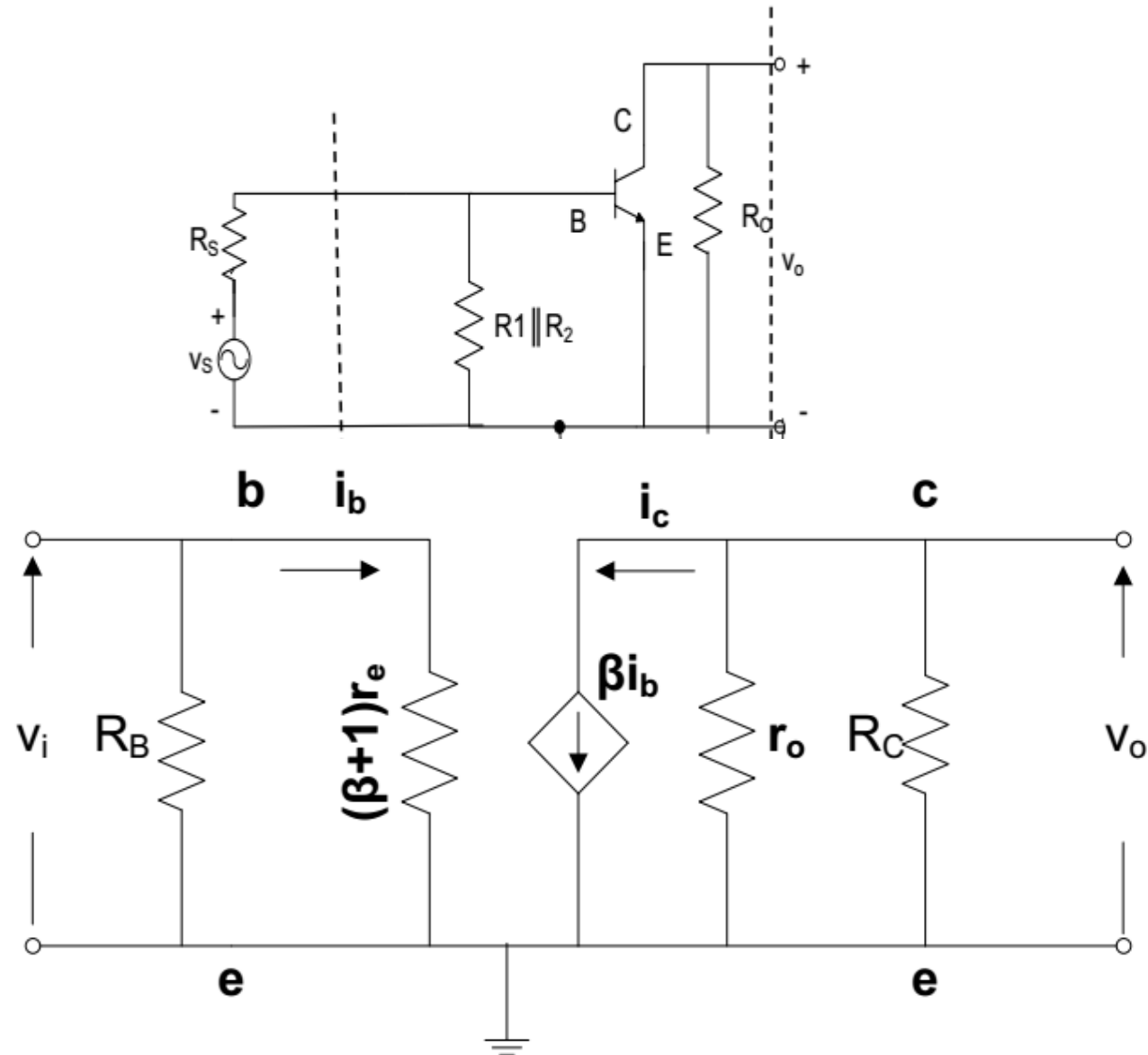
We now replace the transistor with its *small signal equivalent circuit*.
The details of this would be available from the manufacturer's specifications for the transistor.

Add r_o to represent the
output impedance of
the transistor



Using the r_e small signal model (for AC signal analysis)

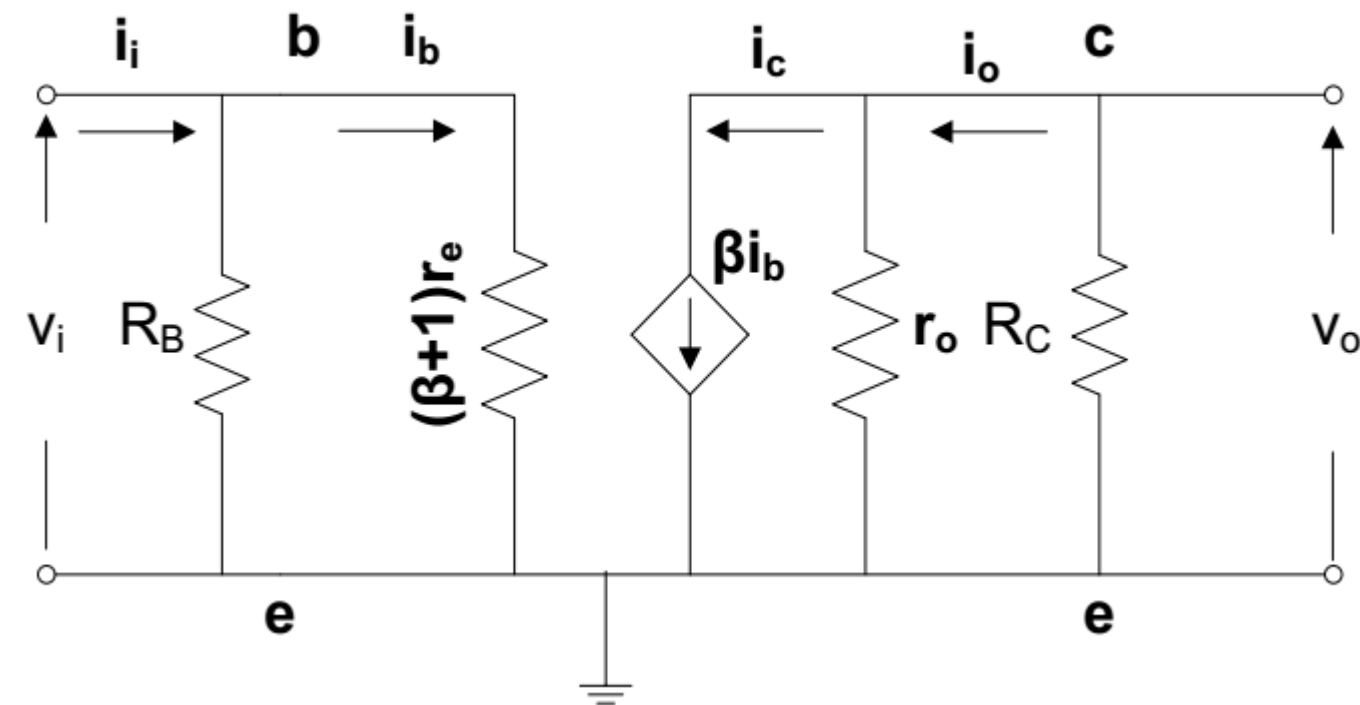
(Without Source and Load Resistances)



$$i_b = \frac{v_i}{(\beta + 1)r_e}$$

$$\frac{v_o}{r_o \parallel R_C} = -\beta i_b$$

$$A_v = \frac{v_o}{v_i} = -\left(\frac{\beta}{\beta + 1}\right) \frac{R_C \parallel r_o}{r_e}$$



For $r_o \gg R_C$ and $\beta \gg 1$ $A_v = -\frac{R_C}{r_e}$

Input Impedance $Z_i = \frac{v_i}{i_i} = R_B \parallel (\beta + 1)r_e$

Output Impedance

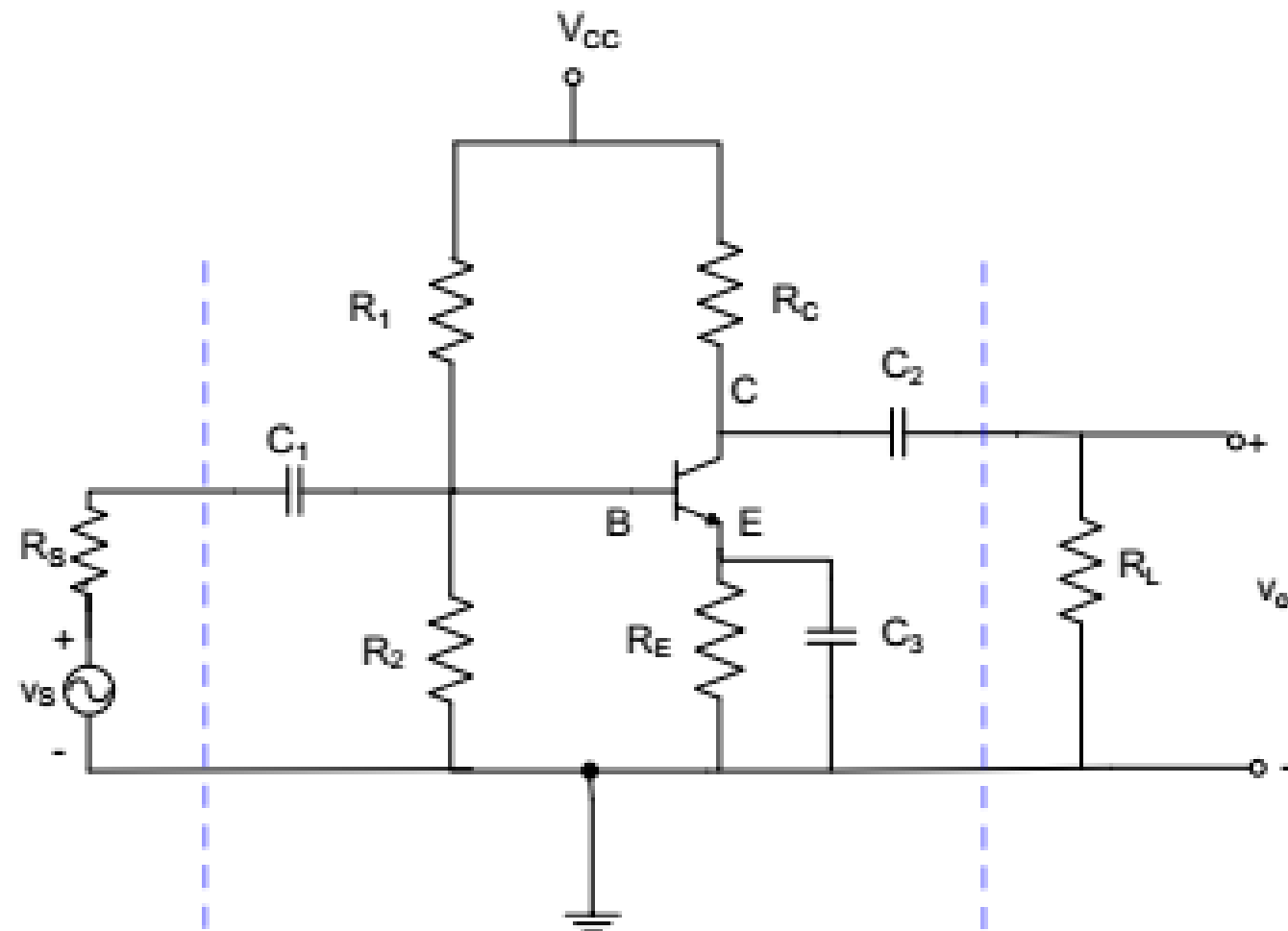
$$Z_o = R_C \parallel r_o$$

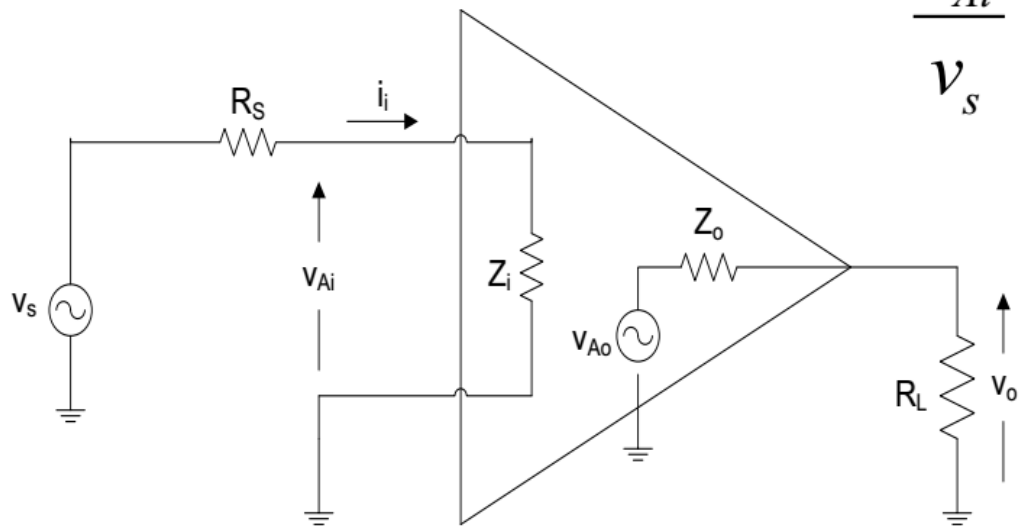
$$i_b = i_i \frac{R_B}{R_B + (\beta + 1)r_e}$$

$$i_o = \beta i_b \frac{r_o}{r_o + R_C}$$

Current Gain $A_i = \frac{i_o}{i_i} = \beta \left(\frac{R_B}{R_B + (\beta + 1)r_e} \right) \left(\frac{r_o}{r_o + R_C} \right)$

Basic BJT Amplifier (with source resistance R_S and load resistance R_L)



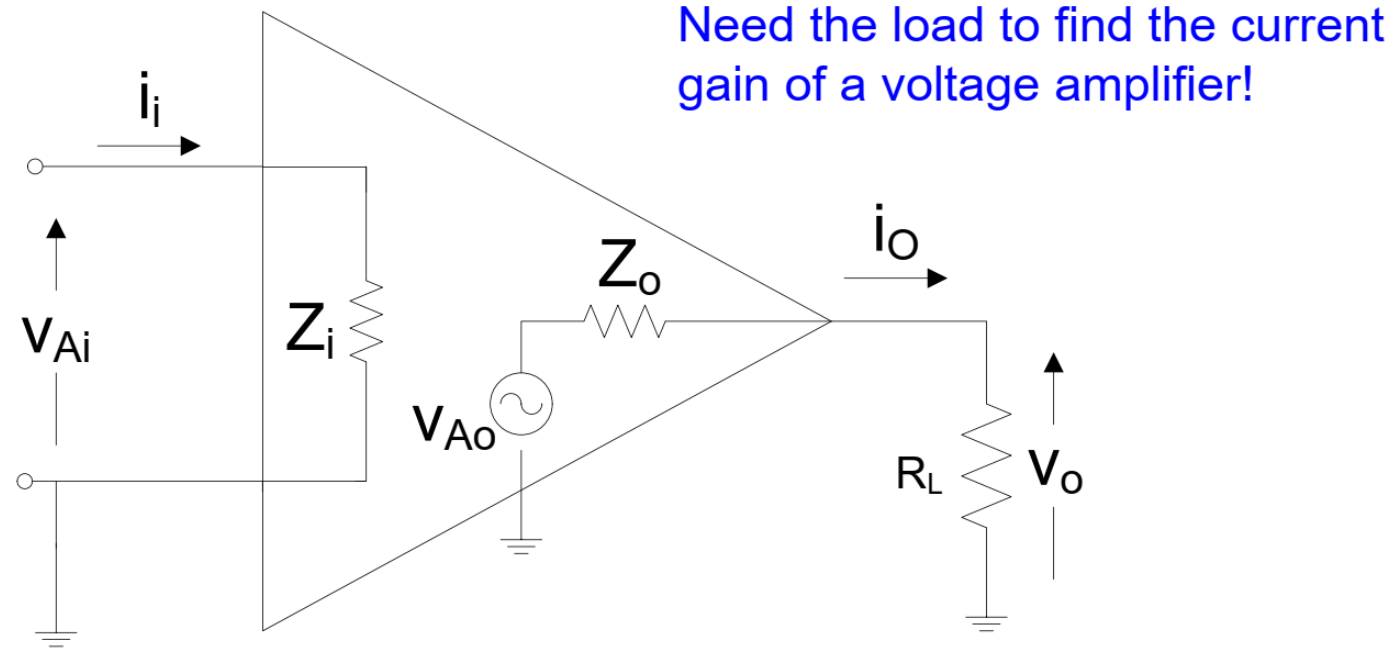


$$\frac{v_{Ai}}{v_s} = \frac{Z_i}{Z_i + R_s}$$

$$\frac{v_o}{v_{Ao}} = \frac{R_L}{R_L + Z_o}$$

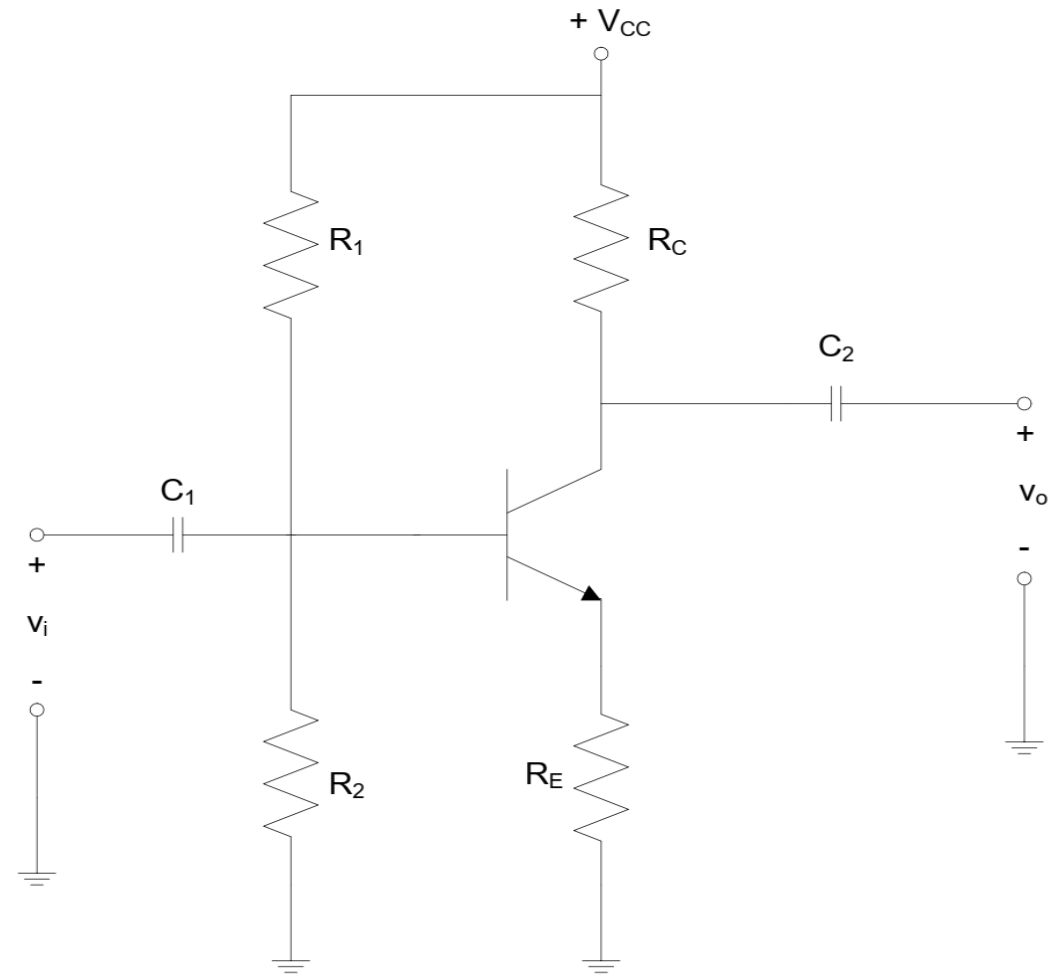
$$\begin{aligned} A_v^* &= \frac{v_o}{v_s} = \frac{v_o}{v_{Ao}} \frac{v_{Ao}}{v_{Ai}} \frac{v_{Ai}}{v_s} \\ &= \frac{R_L}{R_L + Z_o} A_v \frac{Z_i}{Z_i + R_s} \end{aligned}$$

Relationship between A_v , A_i , Z_i and Z_o

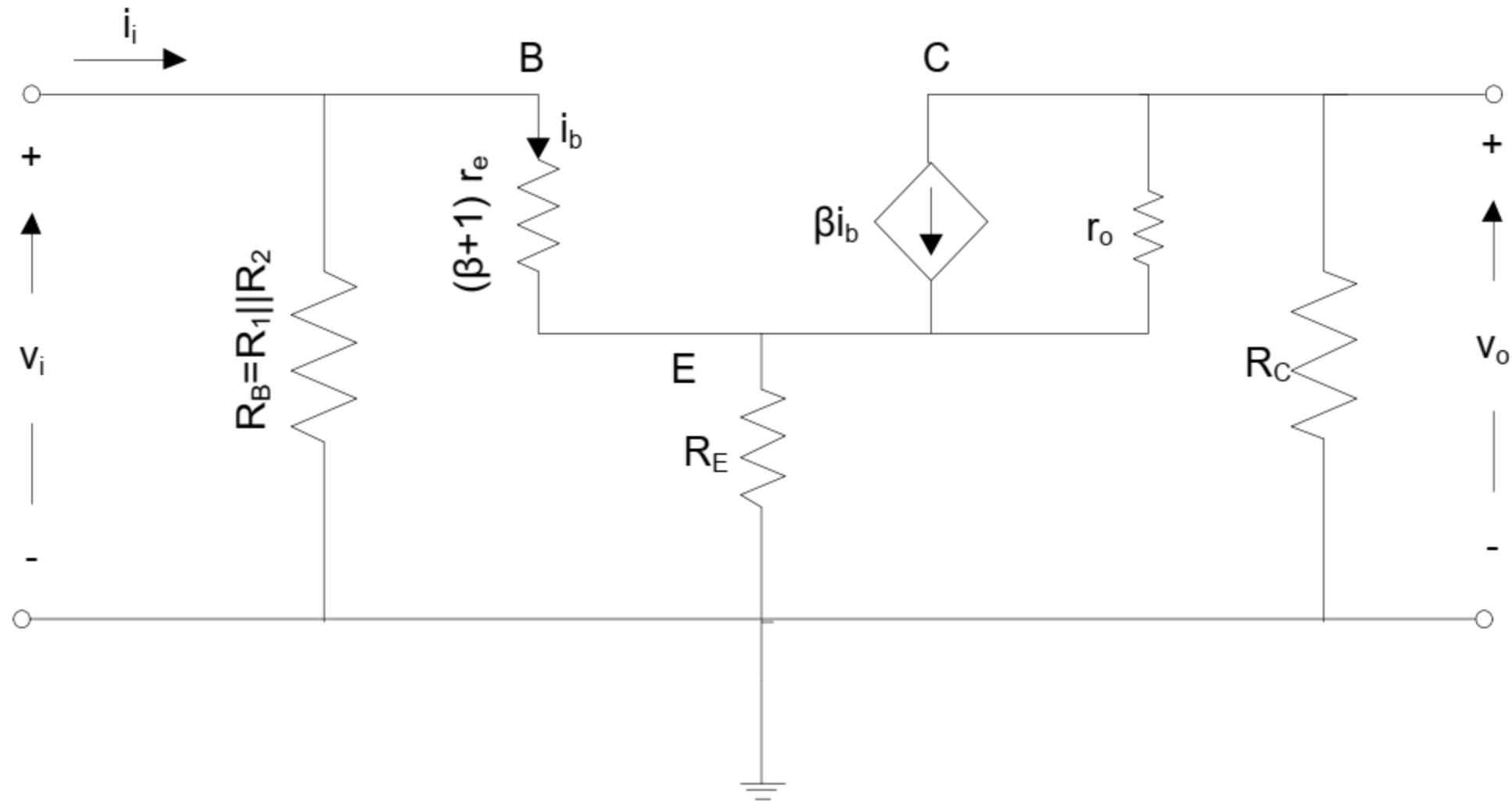


$$\left. \begin{aligned} A_v &= \frac{v_{Ao}}{v_{Ai}} & Z_i &= \frac{v_{Ai}}{i_i} \\ i_o &= \frac{v_{Ao}}{Z_o + R_L} = \frac{A_v v_{Ai}}{Z_o + R_L} = \frac{A_v Z_i i_i}{Z_o + R_L} \end{aligned} \right\} \Rightarrow A_i = \frac{i_o}{i_i} = \frac{A_v Z_i}{Z_o + R_L}$$

Amplifier with Un-bypassed Emitter Resistance



Amplifier with Un-bypassed Emitter Resistance



$$v_i = (\beta + 1)i_b r_e + (\beta + 1)i_b R_E \quad v_o = -\beta i_b R_C \quad \text{Neglecting } r_o$$

Voltage Gain

$$A_V = \frac{v_o}{v_i} = -\left(\frac{\beta}{\beta + 1}\right) \frac{R_C}{r_e + R_E} \cong -\frac{R_C}{r_e + R_E} \cong -\frac{R_C}{R_E}$$

Note that voltage gain reduces compared to the circuit where the emitter resistance has been bypassed using a capacitor!

Output Impedance

$$Z_O = R_C \quad (\text{neglecting } r_o)$$

$$v_i = (\beta + 1)i_b r_e + (\beta + 1)i_b R_E \Rightarrow \frac{v_i}{i_b} = (\beta + 1)(r_e + R_E) = Z_{BE} \quad (\text{say})$$

Input Impedance

$$Z_i = R_B \parallel Z_{BE} = R_B \parallel [(\beta + 1)(r_e + R_e)]$$